MANAGMENT PORTFOLIO AERIS
2025 WORLD FINALS



OVERVIEW

Our participation in the 2025 STEM Racing World Finals is the core project of **Team Aeris**. This project requires rigorous and precise organization to help us achieve our **goals** and showcase our full potential. It is therefore essential to ensure strong **project management** at every stage. To do so, we relied primarily on the *Project Management Guide – 2024/25* provided by the STEM Racing organization, and structured our project around **four main phases**.



KICK OFF MEETING & PROJECT CHARTER



Figure 1: Kick off meeting with Aeris Team members

To ensure effective project management, we started with a kick-off meeting. We produced a **scope statement** that outlined our **goals**, the conditions required to achieve them, and the necessary resources. We also defined the project's **deliverables** and **stakeholders**. Finally, we reviewed the **resources available** to us during the initial phase of the project, along with the associated **risks**. All of these elements discussed during the kick-off meeting were included in the **project charter**, which was initially accessible to all team members and later shared with some of our stakeholders.

SCOPE STATEMENT

To establish the scope statement, we defined three objectives, ranked from the most global to the most personal:

- **Performance objective:** Deliver the highest possible level of performance, both physically during the final (verbal presentation, race, etc.) and through our deliverables (pit display, portfolios, renders, engineering drawings, car, etc.).
- **Community objective:** Build a strong image of Aeris with the support of our followers on social media and our partners, introduce the project to other young people and grow our community by sharing our journey, and offer good media exposure for our team and partners.
- **Personal objective:** Develop technical knowledge and cross-disciplinary skills (project management, communication, etc.) that are essential in the professional world, and strengthen our career path.

Then, for each of these **objectives**, we identified the **conditions** and **resources necessary** for their success, as well as the **learnings** we should gain, as follows:

Scope statement

1. Performance objective

Necessary resources:
- 27 000 €
- a manufacturing partner

Conditions required:
- deliver the deliverables on time
- build a strong team identity
- develop our community
- engage partners and offer them visibility

Learning outcome:
- technical and teamwork skills
- reinforcement of our career choices

DELIVERABLES

Each deliverable must meet the **acceptance criteria**, which are usually set by the competition rules and STEM Racing's technical regulation, before it can be submitted.

The deliverables required for this project:

- Car and their components
- Verbal presentation
- Technical Drawings, Renderings and Portfolios
- Pit Display

STAKEHOLDERS

We then identified our stakeholders during the initiation phase. Stakeholders are all the **individuals** or **groups** who have an **interest** or **influence** in the project.

Stakeholders	Teachers	Sponsors	STEM Racing HQ	School team alumnus	Judges
Stake	Educational monitoring, travel support, guarantee of the school environment.	Funding, visibility, brand image.	Regulations, organisation, validation of deliverables.	Experience and practical advice, ad hoc support.	Fair assessment, methodological rigour, compliance with regulations.
Action	Regular meetings, progress reports, communication via WhatsApp.	Project presentations, creation of sponsorship files, sending of impact reports.	Strict adherence to deadlines, contact via coordinators, file and format compliance.	Request for mentoring, public acknowledgements, invitation to follow our progress.	Carefully prepared deliverables, compliance with regulations, anticipation of potential issues.

Figure 2: Stakeholders Stake and Action identification

It's important to classify them so we can understand their impact and manage our **communication** effectively. That is why we have created this chart, which represents our stakeholders according to their level of **engagement** and **decision-making power**. The higher up on the right-hand side of the chart they are, the greater their impact on our project will be.

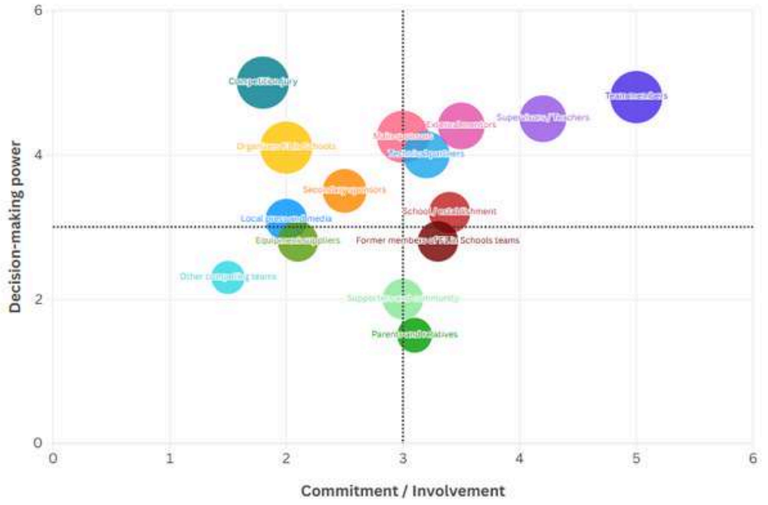


Figure 3: Graphic showing the impact of stakeholders

TYPE OF RESOURCES AND RISKS

Finally, we made sure to consider the **resources** we initially had available, so we could plan for those we needed to acquire. We categorized them into **three groups** and anticipated the risks that could append to each one.

FINANCIAL

Funding is an essential lever for the success of the project. **Costs** related to registration, equipment, travel, and communication must be covered. The major risk is an insufficient budget or poor allocation of funds, which could **block** certain key stages of the project.

TIME

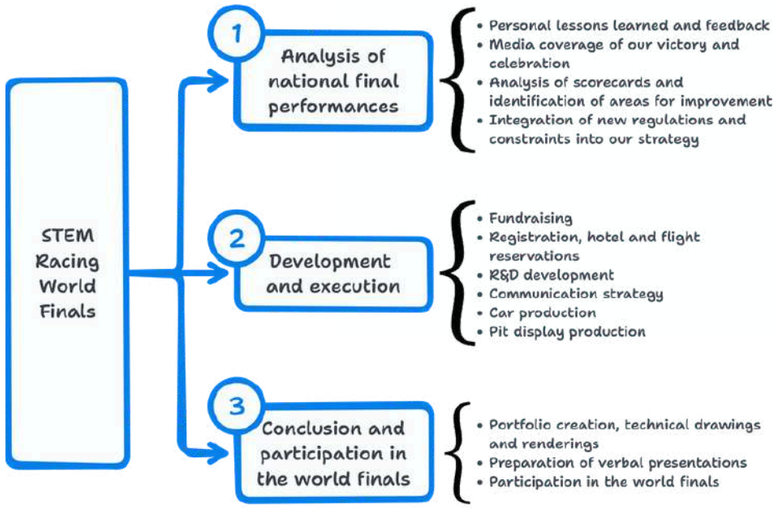
Time is a precious and limited resource. Each team member has a different **schedule** to consider. In addition, we also depend on the limited **availability** of our partners, particularly in August for production.

HARDWARE AND SOFTWARE

We use 3D modelling and CFD simulation software, as well as **equipment** such as the school's 3D printers. Their availability, which is sometimes limited in terms of time, can **restrict our ability to progress efficiently**.

WORK BREAKDOWN STRUCTURE

We divided the STEM Racing World Finals project into three main parts, which allowed us to structure all the tasks more **clearly**. This approach made it easier to allocate **responsibilities** and accurately estimate **deadlines** and **resources**.



PROJECT MANAGEMENT TOOLS

To structure our project and ensure effective monitoring of tasks and deadlines, we compared several **project management tools** that include a **Gantt chart**. Each platform has its own strengths and limitations in terms of functionality, usability, and adaptability to our workflow. Here is an overview of the advantages and disadvantages of the tools we considered:

ASANA

Asana is a widely used project management tool, offering a **clear interface** and a good balance between simplicity and functionality. Its Gantt chart, although practical, **lacks some advanced options**.

MONDAY

Monday offers visual and **interactive** project management with a well-designed Gantt module. It is appreciated for its **flexibility**, but can be **costly** depending on your needs.

NOTION

Notion is a **versatile** platform that allows you to organise databases, centralise documentation and manage tasks. However, its Gantt chart module is **limited** in terms of advanced features.

CLICKUP

ClickUp is a **feature-rich** tool that offers a powerful Gantt chart, ideal for managing complex projects. However, its wide range of features can be **confusing** for new users.

We chose ClickUp for its **balance** of advanced features, **flexibility**, and **accessibility**. Unlike Notion, whose Gantt tools are limited, ClickUp allows for **detailed task** and **dependency management**. Asana, while easy to use, lacked customization for our project tracking needs, while Monday, despite its intuitive interface, was too expensive for our requirements. ClickUp offered us a comprehensive and free solution, ideal for effectively planning our project, tracking deadlines, and adjusting our organization in real time.

TASKS QUALITY ASSESMENT

In order to ensure the relevance and clarity of the assigned tasks, we defined a set of **closed questions** that we had to be able to answer positively before any validation. This systematic verification ensured that each task was **useful**, **well-defined**, **achievable**, and aligned with the project's objectives.

List of questions used:

- Is this task necessary? Does it contribute to one of our scopes?
- Are all the specifics and conditions announced?
- Has a deadline been set?
- Has the RACI been distributed?
- Have the associated risks been identified?
- Is the necessary budget available?
- Have the quality acceptance criteria and quality measurement tool been assigned?

PROJECT SCHEDULE

The **Gantt chart** is a key tool for project management, allowing you to visualise tasks, their **durations** and **interdependencies**. For the STEM Racing project, it was used to plan the different stages, monitor progress and **optimise resource allocation**, thus ensuring efficient management and adherence to **deadlines**.

We structured our Gantt chart on two levels:

- first, in a **general way**, to anticipate all tasks from the project initiation phase
- then, in more **detail**, to manage the execution of tasks on a weekly basis

Thanks to ClickUp, we were able to prioritize elements into tasks and subtasks. This structure gave us greater **flexibility** and more **precise** control over the major stages of the project.

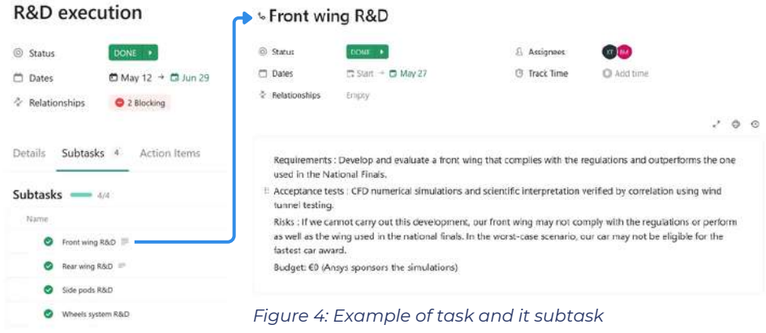


Figure 4: Example of task and it subtask

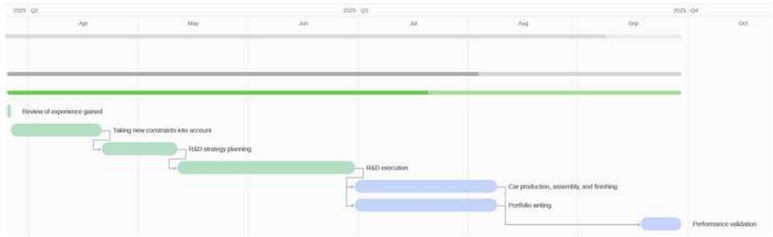


Figure 5: Gantt chart for the Engineering Department

EISENHOWER PRINCIPLES

We used Eisenhower's principle to **prioritise tasks** more effectively and ensure that each team member knew which tasks needed to be done first. We integrated a standard Eisenhower matrix template directly into our ClickUp project management tool. As it was originally designed for an individual, we customised it to work for a team. Each week, we **reviewed the upcoming tasks** and categorised them according to **urgency and importance**. Depending on the category, there were four possible actions.

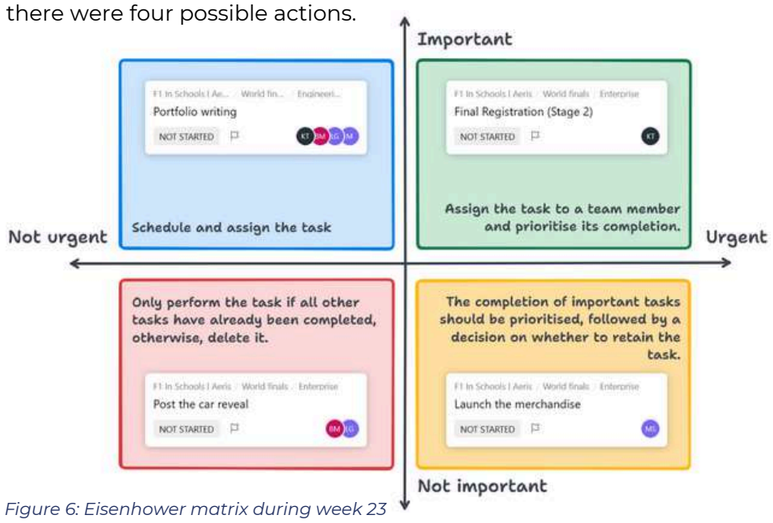


Figure 6: Eisenhower matrix during week 23

RUSH

We planned **two key rush periods**, in order to target the most critical phases of the project :

- The first, in **March**, immediately after our victory, allowed us to increase our media visibility and approach local enterprises.
- The second, scheduled for **July**, took into account the team's unavailability in August and September. This final push was essential for finalizing the deliverables.

QUALITY GATES

Quality Gates are key stages in the project process. After each main task, the **quality** of the task's completion and its **compliance** are assessed before moving on to the next task. They enable the validation of **intermediate objectives**, the detection of potential deviations or problems, and ensure rigorous control throughout the project.

PLANNING

The planning phase is essential for **structuring the project** before its launch. It involves defining objectives, assigning roles, establishing a schedule, and anticipating resource requirements.

ROLES

In order to ensure fluid organisation and efficient task distribution, it was essential to clearly define each person's role. This allowed us to optimise our **collective work** while respecting the **skills** and **abilities** of each member. We took care to develop precise descriptions for each role, based on the needs of the project and the **specific requirements** of each area. This approach enabled us to structure our actions and ensure that everyone knew clearly what they had to accomplish.



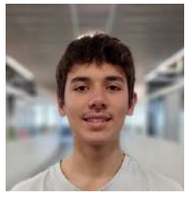
Kaéna TRENCHANT
Roles: Team Leader and CAD Engineer
Missions: Coordinate the team, track tasks and deadlines, facilitate internal communication, design 3D parts, optimize mechanical resistance, prepare files for simulation and manufacturing



Maxime SUTTER
Roles: Sponsorship/Partnerships Manager
Missions: Identify potential partners, manage outreach and negotiations, ensure visibility for sponsors, maintain long-term relationships



Matteo BIETH-DARSTEIN
Roles: R&D Engineer
Missions: Conduct technical research, develop and test innovations, optimize aerodynamics, validate performance through simulations



Lévi GOLDSCHMIDT
Roles: Social Media and Manufacturing Manager
Missions: Management of social networks (content creation, animation, promotion of success stories) and supervision of production (prototyping, quality, final production).

To enable everyone to **widen their knowledge** in various fields, we occasionally **rotated roles**. This allowed Lévi, our Social Media Manager, to contribute to R&D with Mattéo by building a wind tunnel, thereby familiarising himself with design skills and aerodynamics concepts. Similarly, Matteo was able to expand his design skills by contributing to the development of our website.

RESPONSIBILITIES

A clear division of **responsibilities** is essential to ensure efficiency and coordination within a team. The **RACI matrix** (Responsible, Accountable, Consulted, Informed) is a management tool that allows you to precisely define each member's role in the various tasks of the project, thus avoiding confusion and duplication of work.
It operates on four levels of responsibility:

- **R** Responsible (R): Person in charge of executing the task.
- **A** Accountable (A): Person who validates and makes final decisions.
- **C** Consulted (C): Person sought for their expertise or opinion.
- **I** Informed (I): Person kept up to date on the progress of the task.

Thanks to this matrix, we were able to clarify everyone's roles, improve our organisation and optimise communication within the team.

Deliverables / Activities	Maxime	Lévi	Kaéna	Matteo
CAD	I	I	R	C
R&D	I	I	A	R
Production	I	I	A	C
Artistic Direction	R	C	A	I
Marketing	C	R	A	I
Finance	R	I	A	I
Pit Display	R	C	A	I
Entreprise Portfolio	R	R	A	I
Engineering Portfolio	I	I	R	R
Project management Portfolio	A	I	R	C

Entreprise Engineering Project management

Figure 1: RACI Matrix

INTERNAL COMMUNICATION

To ensure smooth organisation and effective collaboration, we used several **communication tools** tailored to our needs. Each one had a specific role, allowing us to centralise exchanges, structure the work and monitor the project's progress in real time.

CLICKUP
Features: Planning, deadline management, project visualisation
Frequency of use: Weekly

NOTION
Features: Document management, task tracking, resource organisation
Frequency of use: Daily

GOOGLE DRIVE
Features: Cloud storage, live collaboration, smart sharing, version tracking
Frequency of use: Daily

DISCORD
Features: Voice meetings, themed discussions, real-time collaboration
Frequency of use: Weekly

WHATSAPP
Features: Instant messaging, announcements, quick coordination
Frequency of use: Daily

We also organised a **weekly meeting** to keep all members informed of the project's progress. It was held in person at the school on Wednesdays when we had classes (every other week), or remotely on Discord the other weeks.

LINK ENTERPRISE AND ENGINEERING

Effective **communication** between the engineering and business departments is essential to ensure consistent and aligned progress within the team. That is why we have implemented a structured system for **accessing technical files**, allowing members of the business team to access essential information without compromising the integrity of technical designs. Onshape is used to share design files, with access rights assigned according to need. In addition, Google Drive centralises and organises non-technical documents, such as marketing and administrative materials, making them easier for different departments to consult. In order to assign these access rights in a consistent manner, we have developed a **decision tree** to determine whether a user should be granted edit, read-only or comment rights, thus ensuring a balance between **accessibility and data security**.

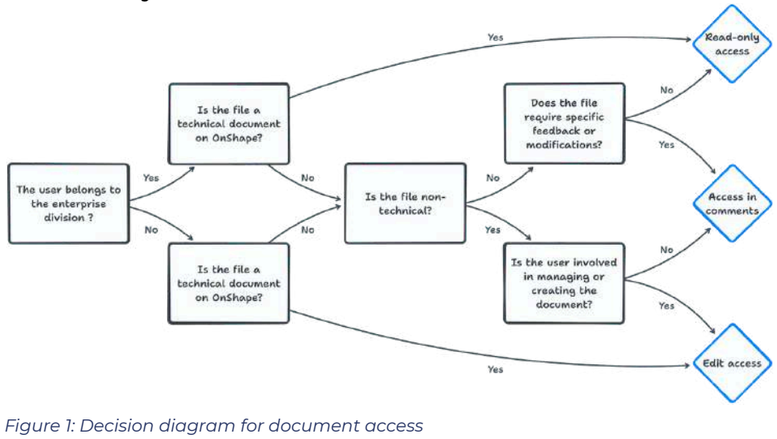


Figure 1: Decision diagram for document access

FINANCIAL BUDGET

After winning the French national final, we began estimating the budget needed for the world final. Fundraising is a step that must be **anticipated**, especially when it represents our biggest obstacle to reaching the world final. That is why estimating the budget is a key step in planning a project. It allows us to anticipate the amount of funds to be raised, guide spending **strategically**, and ensure the project's **success**.

As the project planning progressed, the number of participants and the prices of various services (airline tickets, exchange rates, etc.) changed, which led us to regularly **adjust** the budget. In the end, our team consisted of five members and two accompanying adults.

	Value
Hotel and Entry Fee	17 464,32€
Travel	4 400€
Car production	400€
Pit Display	1 000€
Total	23 464,32€

Figure 1: Initial budget prognosis

To estimate our budget, we were fortunate to benefit from the expertise of a former team from our school: **Dynamis Racing**. They themselves had participated in the 2023 F1 in Schools World Finals in Singapore. We anticipated costs for various categories, such as hotel and participation fees, transport costs, car production and pit display. In addition, we anticipated **deadlines** for each of these expenses (in a more detailed Google Sheets table). For example, we had to pay the deposit within 30 days of 25 April.

In retrospect, we noticed a **difference** between the estimates and the actual hotel and participation costs. These costs could have been estimated accurately, as the rates had been communicated in advance. However, **exchange rate** fluctuations, which are difficult to predict, led to a noticeable but minimal difference. Regarding travel, airfare prices turned out to be higher than expected, mainly because we booked them a bit late, due to **funds** arriving gradually.

	Value
Hotel and Entry Fee	17 257,21€
Travel	4 630,83€
Car production	332,98€
Pit Display	---€
Total	22 221,02€ + ---€

Figure 1: Final real budget

The cost **variations** between estimated costs and actual spending in certain categories had a direct **impact** on other budget items. In order to maintain the overall balance of the budget, we therefore decided to **reduce** the spending allocated to the production of the car and the manufacture of the pit display.

INCOME TRACKING

A To meet the **requirements** of this budget, we managed to raise over 24 000€ through sponsorship, by reaching out to our former partners and also by convincing new sponsors to join us in this adventure.

	Value		
Rotary Club	5 000€	Helbling	1 072,02€
Crédit Mutuel Blotzheim	3 000€	Sofitex	1 000€
Saint-Louis Agglomération	3 000€	Pierre Goldschmidt	1 000€
Lions Club	2 500€	Valoria	500€
Fundraising on Tribee	2 170,75€	Mairie Biederthal	500€
Alsace Dépannage	1 800€	Productec	490€
		Guera Construction	250€
Total	24 782,77€		

SPONSORING INVOICES



As a **non-profit organisation** governed by the French law of 1901, we were able to offer our French partners a tax **deduction of up to 66%** of the donation amount, within the 20% tax reduction limit. This was a strong selling point for us when fundraising. However, to do so, we had to put in place rigorous administrative procedures, including issuing **invoices for sponsorships** and carefully tracking our income and expenses.

Figure 1: Example of invoice for sponsorship

EXPENSE TRACKING

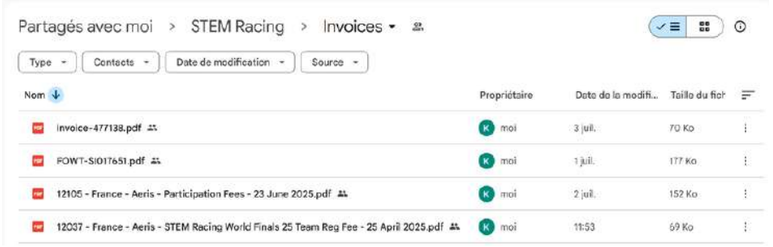
In order to **accurately track** our budget and ensure complete transparency in the management of funds, we decided to open a dedicated **bank account** in the name of our association. This account was opened with Crédit Mutuel de Blotzheim, a partner bank located near Saint-Louis, the town where our school is located. Thanks to this account, we had complete **autonomy** over the management of financial flows: receiving sponsorship transfers, making payments to service providers, monitoring the balance in real time, verifying transactions, etc. The management of this account quickly became an essential tool for steering the project.



Figure 1: Screenshot extract from our banking app

EXPENSE ARCHIVE

In the interests of transparency and organisation within the team, we set up a shared space on Google Drive containing all the essential project documents, including **invoices**. This allowed each member to access all financial information at any time, avoid losing documents, and facilitate **communication** in the event of an audit or request from our sponsors. This simple but effective system strengthened confidence in the collective management of the project, without requiring complex software or special skills.



FINANCIAL CLOSING

We have also planned the **financial closing** of the project. After the competition, we will finalise the expense receipts and record the donations made by our partners in order to draw up the **final financial statement** for the project. We also plan to reinvest any surplus funds in thanking our sponsors.

The Executing phase corresponds to the actual **implementation of the project**. It involves coordinating actions, mobilising stakeholders and adapting to risks.

STAKEHOLDER ENGAGEMENT MATRIX

We created a Stakeholder Engagement Matrix to determine the level of **involvement** required for each project stakeholder. Based on two criteria, their **interest** in the project and their **influence** on its progress, we were able to decide whether to simply keep them informed, consult them on an ad hoc basis, or involve them closely in certain decisions.
We also identified each stakeholder's main area of involvement (engineering or enterprise), which allowed us to tailor the content of our exchanges and better target moments of interaction.
This approach enabled us to optimise our coordination, focus our efforts in the right areas, and **maintain stakeholder engagement** throughout the execution phase.

Stakeholder	Domain	Influence	Interest	Engagement strategy
Additional Team members	Enterprise and Engineering	High	High	Active Get involved actively
Principal financial sponsors	Enterprise	High	High	Manage closely
Technical sponsors	Engineering	High	High	Get involved occasionally
Our School	Enterprise and Engineering	High	Middle	Keep informed
Acbcom	Enterprise	Middle	Middle	Get involved occasionally
Dynamis Racing and Solarity Racing team members	Enterprise and Engineering	Low	Middle	Keep informed
Other minor sponsors	Enterprise	Low	Middle	Keep informed

Figure 7: Stakeholder engagement matrix

EXTERNAL COMMUNICATION PLAN

We implemented a structured external communication plan with our stakeholders to precisely define contact dates, the information to be communicated, and the means of communication to be used. This planned follow-up enabled us to maintain good **coordination** with our partners throughout the project.
This was essential for meeting **deadlines**, particularly during critical production phases. By anticipating each exchange, we were able to ensure that all information was communicated on time, **clearly** and **completely**, and that each partner had the necessary elements to contribute effectively to the progress of the project.

Who ?	What ?	Method ?	When ?
Helbling	Documents containing technical data and calculation requests	Mail	4th January
ISL	STLs of our wheels, material, technical drawings	Mail	15th June
SunChemicals	Car delivery and hex code of our colours	Google Meeting and Google Docs shared	15th June
Acbcom	Merch design, polos design, etc	Mail	5th April and 25th July
Others	Thank you message	Mail and WhatsApp	8th October

Figure 8: External communication plan

SPONSORS MEETINGS

We have welcomed our partners on several occasions in order to maintain a **lasting relationship** of trust with our stakeholders and to ensure their **continued commitment** between the national and world finals. These exchanges have also helped to advance the project, drawing on their specific expertise.
For example, a meeting was organised at the beginning of the year with the **ISL (Franco-German Research Institute in Saint-Louis)** to enable the entire team to better understand the concepts of aerodynamics, which were entirely new to us at the time.



Figure 9: Meeting with SunChemicals



Figure 10: Baptiste Masse, who gives us a brief lesson on aerodynamics

POTENTIAL RISK MANAGMENT

Any ambitious project involves uncertainties that can impact on its progress and results. **Proactive risk management** is therefore essential to anticipate obstacles and limit their consequences. Identifying potential threats helps to optimise decision-making and ensure continuity of work without major disruption. To do this, we have put in place a structured **risk analysis**, assessing the likelihood of occurrence and the impact on the project. This approach enables us to define appropriate mitigation strategies and act quickly in the event of unforeseen circumstances.

To make this analysis more effective and visual, we use a risk matrix, which ranks and prioritises the different challenges we could face.



Figure 11: Risk matrix

Let's take risk no. 2 identified in our matrix as an example. If this risk materialises, the consequences would be **critical**, as it could prevent us from assembling our car on time, thereby compromising our participation in the races. To **limit this probability**, we have taken precautions by finding out our partner's schedule and setting a clear deadline for the transmission of our files and technical plans. If, despite these precautions, the risk were to occur, we have considered an **alternative solution** by identifying other potential partners who could take over in an emergency. This pre-planned responsiveness would allow us to limit the impact of the problem and ensure the continuity of the project.

UNEXPECTED RISK MANAGMENT

Despite all the precautions taken, unexpected risks can always arise. That is why we have put in place an **emergency procedure** to enable us to respond quickly and effectively. In the event of a critical situation, the steps to follow are as follows:

1. **Identify** the nature of the risk precisely.
2. **Inform** the entire team immediately.
3. The Team Leader **assesses** and **classifies** the risk according to its severity.
4. If the risk is critical, an **emergency meeting** is organised on Discord with the available members. Otherwise, the item is added to the agenda for the next weekly meeting.
5. **Discuss** possible solutions collectively.
6. **Make a decision**, ideally by consensus. In the event of disagreement, a vote may be organised.
7. **Act** quickly to limit the consequences.

MONITORING

Finally, the monitoring phase allows you to **track the progress** of the project in real time, measure deviations from the initial plan, and react quickly in the event of unforeseen circumstances.

QUALITY MANAGEMENT

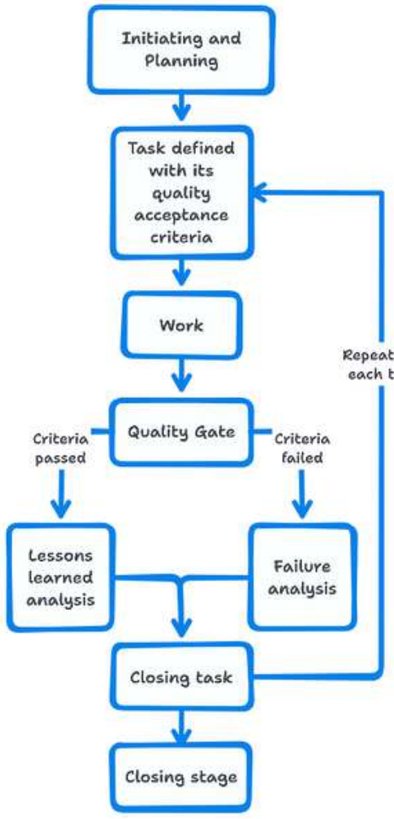


Figure 12: Quality management cycle

To guarantee the quality of execution at every stage of the project, we have implemented a **structured quality management process**. After each important task is completed, its **compliance** is assessed using measurement tools and criteria defined during the planning phase (see Initiating > Tasks Quality Assessment). These **criteria** vary depending on the nature of the task: part dimensions, machining accuracy, visual rendering, or compliance with specific specifications.

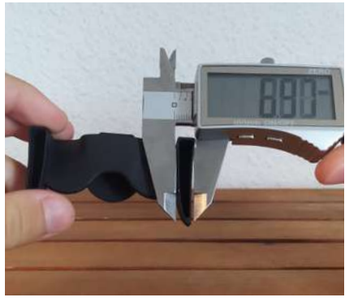


Figure 13: Front wing measurement

When the criteria are met, we validate the task and analyse the lessons learned to improve our methods. However, if the criteria are not met, we analyse the causes of failure (technical, organisational or human) before correcting or repeating the task.

STATUS REPORT

When managing a project, the first thing to monitor is, of course, the **progress of tasks**.

In our case, we wanted to simplify this process. Each team member now has direct access to their tasks on ClickUp, and everyone can **update the status** of their assigned tasks in real time. ClickUp also offers a **dashboard** that provides an overview of the progress of assigned tasks.

This was very useful during our weekly meetings, enabling us to monitor our progress and keep all **project members informed**.

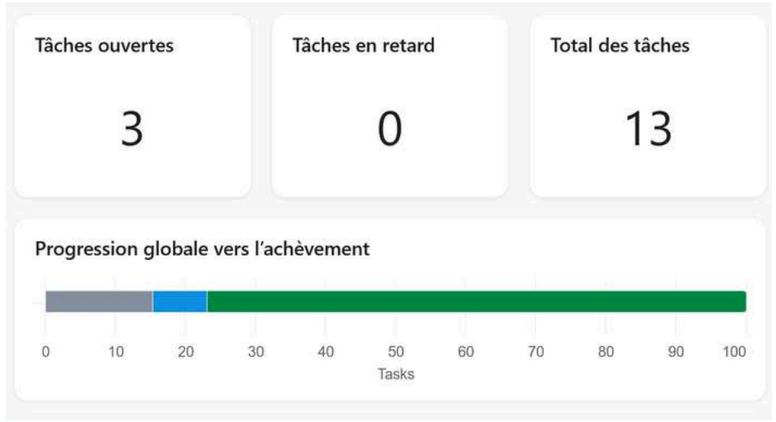


Figure 14: Engineering Dashboard

This enabled us to measure our progress **qualitatively**. For long tasks, we conducted a **weekly review** of the task status, including the subtasks to be completed, those in progress and any problems encountered.

ANALYSIS AFTER FAILURE

After identifying a task that does not meet the defined quality criteria, we conduct a **detailed analysis** of this failure. This allows us to better understand the cause of the issue, so that we can **correct** it and **prevent** it from happening again.

To do this, we ask ourselves a series of **targeted questions**:

- What is the impact of this failure ?
- What critical decisions, actions, or miscommunications led to the failure?
- Which ways do we have to solve this issues ?
- What are the consequences on ressources ?
- Are sponsors involved ?

QUALITY IMPROVEMENT

In order to measure improvements in task quality, we have implemented a monitoring system based on **counting** compliant tasks. This system has also enabled us to **adjust the questions** used in failure analysis, making them more relevant. The **radar chart** opposite shows an increase in the percentage of tasks completed satisfactorily between the National Finals and the World Finals.



This improvement can be explained in particular by the **revision of our method** of analysing failures between these two events.

LESSONS LEARNED

After each task, all team members discuss the lessons learned, but it is especially after each competition that we take stock of what F1 in Schools and STEM Racing have taught us.

To structure this reflection, we have defined two areas of analysis:

- The first, the **collective** area, brings together all the lessons learned relating to the team as a whole. These lessons are directly reusable and enable us to continuously improve the progress and coordination of the project.
- The second, the **personal** axis, brings together the **technical** and **cross-disciplinary skills** that each member has developed, which will be valuable for their individual development.

For example, during the National Finals, after presenting the project to the judges, the team strengthened its ability to communicate effectively under pressure, manage stress and respond to unexpected questions.
On an individual level, Matteo improved his technical skills in computational fluid dynamics (CFD), while developing cross-functional skills such as pitching to sponsors and project management.

Each of these lessons learned contributed to the **advancement** of the Aeris team's project, but not only that, it also allowed us all to advance in our own **personal projects**.

CLOSING STAGE

After finalising the deliverables, our Aeris team will validate its **performance** at the global finals in Singapore, where their work will be assessed by the competition's expert judges. The competition therefore marks a **key milestone**, initiating the closing stage phase of the project.

The conclusion of the project is also an opportunity to **recognise the work** accomplished by each of us and to thank all those who have been actively involved in making our adventure possible.

Finally, just as we did after the National Finals, we will take the time after the competition to **celebrate** the culmination of our efforts together. We will also organise a convivial gathering of sponsors, families, friends, teachers and members of the school to express our **gratitude** and celebrate the success of this adventure.



Figure 15: Celebration of our victory in the national finals